

SEMINAR III · WEEK 6 · FOUNDATIONS

Ratios, Rates & Proportions

The big question this week: what two quantities am I comparing?

Two teaching sessions, not four separate lessons.

SESSION A

Ratios and Rates

WHAT IS A RATIO?

Part-to-part, or part-to-whole?

students to computers

wins to losses

recipe ingredients

Same relationship, three ways to write it: 3 to 2, 3:2, $3/2$.

COMMON MISCONCEPTION

Does 3:2 mean the same as 2:3?

No — order tells you which quantity is being compared to which.

COMMON MISCONCEPTION

Does simplifying 18:12 to 3:2 change what's being compared?

| *No — same relationship, different scale.*

COMMON MISCONCEPTION

5 forwards : 7 defenders — are forwards 5/7 of the team?

No. The team has 12 players total, so forwards are 5/12 of the team — not 5/7. Part-to-part isn't part-to-whole.

EQUIVALENT RATIOS & UNIT RATES

Scale by multiplying, not adding.

Wrong: "add 2 cups of sugar, so add 2 cups of everything else." Right: multiply or divide every quantity by the same factor.

COMMON MISCONCEPTION

Does a higher price always mean a worse deal?

No — the unit rate determines value, not the total price.

APPLICATION — BEST BUY

Compare two offers using unit rates.

- groceries
- phone plans
- subscriptions

EXIT TICKET — SESSION A

1. Generate an equivalent ratio
2. Calculate a unit rate
3. Pick the better value between two rates

SESSION B

Proportions and Unit Conversion

SETTING UP A PROPORTION

Matching quantities in matching positions.

Introduce cross multiplication and the unit-rate method as two paths to the same answer — then always check the solution.

COMMON MISCONCEPTION

Can any two fractions in a proportion be cross multiplied?

Only if the proportion was set up with matching quantities in matching positions.

COMMON MISCONCEPTION

Is a unit rate always the larger number divided by the smaller?

No — divide so the reference quantity becomes 1, based on what the question asks "per," not which number is bigger.

UNIT CONVERSION

A conversion factor is just a ratio.

The trick is arranging it so the unwanted unit cancels — and checking multi-step conversions with estimation.

COMMON MISCONCEPTION

Does it matter which unit goes on top of a conversion factor?

Yes — flip it and the unwanted unit doesn't cancel. The result comes out wrong, often obviously unreasonable.

ACTIVITY — CONVERT AND COMPARE

Price per ounce vs. price per pound — convert to a common unit first.

- Why does the unit need to match before comparing?
- What happens if the conversion factor is flipped?
- How can estimation catch an unreasonable conversion?

CLOSING THIS WEEK

- Mixed practice: ratios, rates, proportions, conversions
- Error analysis on missed proportion setups, then retry
- Week 6 check, then reflection

TRANSITION TO WEEK 7

Variables & Expressions

A proportion asks us to find an unknown quantity. Algebra gives us a more general way to represent and solve for unknowns.